

GELA, Italy

WMO: 164530

Lat: 37.076N

Lon: 14.225E

Elev: 69

StdP: 100.50

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.5	6.9	-1.1	3.5	9.4	0.8	4.0	9.5	15.1	13.8	13.4	13.5	2.5	30	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	29.7	24.5	28.6	24.4	27.8	24.2	26.5	28.4	25.7	27.7	25.0	27.2	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.4	28.1	25.0	20.3	27.5	24.2	19.3	26.9	83.0	28.5	79.5	27.9	76.5	27.2	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.2	10.3	3.3	32.7	1.9	1.6	1.9	33.9	0.8	34.8	-0.2	35.7	-1.6	36.9		
(5)			2.1	27.6	1.5	1.0	1.0	28.3	0.1	28.9	-0.8	29.5	-1.9	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	12.4	12.0	12.6	14.5	17.9	21.4	23.7	25.2	23.3	20.3	16.3	13.4
	DBStd	5.14	1.85	1.96	2.27	2.13	2.50	2.20	2.05	1.59	1.84	2.16	2.54	2.18
	HDD10.0	18	3	5	6	0	0	0	0	0	0	0	1	3
	HDD18.3	930	184	178	178	117	39	2	1	0	0	8	70	153
	CDD10.0	2861	77	61	87	134	244	343	426	471	400	321	189	109
	CDD18.3	731	0	0	0	2	24	95	168	213	150	69	8	1
	CDH23.3	3810	0	0	0	2	49	259	881	1696	786	136	1	1
CDH26.7	479	0	0	0	0	4	24	112	272	64	4	0	0	
Wind	WSAvg	3.8	3.5	3.9	4.1	4.7	4.0	3.9	3.5	3.4	3.6	3.5	3.8	3.7
Precipitation	PrecAvg	539	80	57	48	36	15	10	5	9	50	63	76	86
	PrecMax	874	225	229	172	85	42	39	29	31	141	132	213	227
	PrecMin	270	3	7	0	0	0	0	1	1	3	0	10	4
	PrecStd	150	53	51	37	23	12	11	6	9	43	36	48	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	18.6	20.4	22.8	26.9	29.0	30.9	31.5	29.6	27.8	23.1	20.0
		MCWB	15.0	13.7	13.7	15.2	18.3	20.3	23.4	25.0	22.7	23.5	19.6	16.5
	2%	DB	17.0	17.0	18.2	20.8	24.8	26.9	29.2	29.9	28.1	25.8	22.0	18.6
		MCWB	13.9	13.4	13.8	15.5	18.8	20.4	24.1	25.5	24.0	22.0	18.7	15.6
	5%	DB	16.2	16.1	17.1	19.2	23.2	25.5	28.1	28.9	27.2	24.9	21.1	17.8
		MCWB	13.3	13.2	13.7	15.2	18.5	21.1	24.0	25.3	23.7	21.7	18.0	15.0
	10%	DB	15.4	15.4	16.2	18.0	21.9	24.7	27.1	28.1	26.5	24.1	20.1	16.8
		MCWB	12.8	12.7	13.4	15.2	18.2	21.1	23.3	24.9	23.4	21.3	17.2	14.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	15.5	16.0	17.3	22.1	24.2	27.2	27.7	26.0	24.6	20.9	17.5
		MCDB	17.5	17.0	18.0	19.6	24.2	26.0	28.8	29.4	27.9	26.4	22.6	19.3
	2%	WB	14.8	14.6	15.1	16.6	20.7	23.2	26.0	26.9	25.2	23.2	19.4	16.3
		MCDB	16.6	16.2	17.0	18.8	22.7	24.9	28.1	28.7	27.0	24.8	21.4	18.1
	5%	WB	14.0	13.9	14.4	16.0	19.7	22.4	25.0	26.1	24.6	22.4	18.4	15.3
		MCDB	15.7	15.7	16.3	18.3	21.7	24.4	27.3	28.1	26.6	24.3	20.6	17.2
	10%	WB	13.2	13.2	13.7	15.4	18.8	21.7	24.1	25.5	24.0	21.6	17.6	14.4
		MCDB	15.0	15.1	15.8	17.6	21.1	23.9	26.3	27.6	26.1	23.8	19.8	16.4
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.9	6.1	5.8	5.4	4.7	4.6	4.7	5.4	5.6	5.6	5.5
		MCDBR	5.8	6.5	7.1	7.2	7.0	6.2	6.0	5.8	6.1	6.1	5.3	6.3
		MCWBR	4.3	5.0	5.5	4.9	5.7	5.7	6.2	6.3	6.0	4.9	4.2	4.7
	5% WB	MCDBR	5.8	5.9	6.4	6.1	5.5	4.8	4.6	5.0	5.7	5.6	4.9	5.6
		MCWBR	4.9	5.1	5.6	5.0	6.0	6.1	6.6	6.3	6.0	5.3	4.2	4.9
Clear-Sky Solar Irradiance	taub	0.351	0.373	0.422	0.463	0.465	0.444	0.446	0.454	0.448	0.422	0.386	0.356	
	taud	2.405	2.322	2.171	2.076	2.118	2.216	2.224	2.201	2.225	2.272	2.329	2.397	
	Ebn at Noon	827	852	837	822	826	841	836	821	803	786	774	792	
	Edh at Noon	89	109	139	161	157	142	140	140	130	113	96	85	
All-Sky Solar Radiation	RadAvg	2.37	3.18	4.47	5.71	6.80	7.55	7.76	6.83	5.17	3.74	2.62	2.13	
	RadStd	0.18	0.30	0.32	0.36	0.35	0.34	0.20	0.33	0.24	0.25	0.19	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.37	-0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page